

CHAPTER 11: INDIVIDUAL VEHICLE CONTROL SIGNALS

11.1 OPERATION

- 1 Traffic signals for the control of individual or single vehicles, as distinct from those that give right of way to groups of vehicles, are used to control traffic at locations such as freeway on-ramps, toll booths and roadside checkpoints.
- 2 The Type S12 traffic signal face is used for the control of individual vehicles as shown in Figure 11.1. The signal face comprises only a RED DISC and a GREEN DISC LIGHT SIGNAL.
- 3 A yellow signal aspect is not provided in the S12 signal face. The signals should therefore not be used to control vehicles other than those that are stationary or travelling at low speed. This can be achieved by:
 - (a) displaying the green signal only to a vehicle that has already stopped at a stop sign, or other similar sign, near to the signal (such as at toll booths and checkpoints); or
 - (b) resting the signal in red and displaying the green signal ONLY when required, and then only for a few seconds to allow one stopped vehicle to depart at a time (such as when ramp metering is applied).
- 4 Where it is required to give continuous right of way to all approaching vehicles, the green light signal may be displayed continuously. When it is necessary to switch the signal to red, a flashing red light signal should first be displayed for a duration of at least 5 seconds.
- 5 At no time SHALL an operational traffic signal be intentionally switched off and blacked out, other than for maintenance or repairs or when controlled by a traffic officer or an authorised pointsman. Flashing red light signals may also be used to indicate that the signals are out of order.

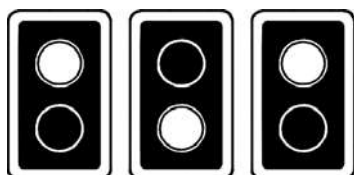
11.2 TOLL BOOTHS AND CHECKPOINTS

- 1 The S12 traffic signal face is used at toll booths and checkpoints to instruct vehicles either to stop or to continue.
- 2 A STOP sign R1, or any other sign that instructs the driver to stop, should be displayed at the stopping point. At least one S12 traffic signal face should then be provided per lane of traffic, located on the right-hand (driver's) side of the lane. The signal face should be located not less than 6 m beyond the stop position.
- 3 The signal should wherever possible, be controlled automatically. The duration of the green light signal and change to red can best be controlled by the output from vehicle detectors in each lane. The illumination of the green light signal may be performed automatically, e.g. linked to a toll booth cash register, or manually.

11.3 RAMP METERING

11.3.1 General

- 1 Ramp metering is applied to restrict the number of vehicles allowed to enter a freeway in order to ensure an acceptable level of service on the freeway or that the capacity of the freeway is not exceeded. The need for ramp metering may arise due to factors such as:
 - (a) Recurring congestion because traffic demand exceeds the capacity of roads in an area.
 - (b) Sporadic congestion on isolated sections of a freeway because of short-term traffic loads from special events, often of a recreational nature.
 - (c) As part of an incident management system to assist in situations where an accident downstream of the entrance ramp causes a temporary drop in the capacity of the freeway.
- 2 Ramp metering should, however, be considered a last resort rather than a first option in securing an adequate level of service on the freeway. Prior to its implementation, all alternative means of improving the capacity of the freeway or reducing the traffic demand on the freeway should be explored.



S12 traffic signal face

Figure 11.1: Single vehicle release operating sequence

- 3 Installation should be preceded by an engineering study of the physical and traffic conditions on the freeway facilities likely to be affected. These facilities include the ramps, the ramp terminals and the local streets likely to be affected by metering as well as the freeway sections involved.
- 4 The study should include the establishment of desirable metering rates. The effect of metering rates on the level of service on the freeway as well as the street network should be evaluated. Attention should be given to storage requirements on ramps and the possible impact of queues on local streets.
- 5 A problem with ramp metering is the need for law enforcement. Without enforcement, infringement rates can be expected to be high. If, for any reason, it is not possible to ensure regular enforcement, ramp metering should not be considered.

11.3.2 Installation

- 1 The stop line should be placed well in advance of the point at which ramp traffic will enter the freeway to allow vehicles to accelerate to approximately the operating speed of the freeway, as would normally be required for the design of on-ramps.
- 2 It will also be necessary to ensure that the ramp has adequate storage to accommodate the vehicles queuing upstream of the traffic signal.
- 3 The above requirement will almost certainly lead to a need for reconstruction of any ramp that is to be metered. The lengths of on-ramps are typically determined by the distance required to enable a vehicle to accelerate to freeway speeds. Without reconstruction, this could result in the ramp metering signal actually being installed at the ramp terminal.
- 4 A STOP LINE RTM1 shall be provided on the on-ramp. At least two S12 traffic signal faces should be provided for ramp metering at a distance not less than 6 m (preferably not less than 10 m) beyond the stop line.
- 5 A FLASHING YELLOW WARNING SIGNAL SS3 could, with advantage, be installed at the start of the ramp to warn vehicles that metering is in operation.
- 6 When ramp metering is in operation, the S1 signals should normally rest in red, and a green light signal displayed ONLY when required and then only for the time required by the departing vehicle to clear the line of vision of the signal face. Such timing should preferably be achieved by means of vehicle detectors. At least two detectors would normally be required for this purpose, namely the check-in and check-out detectors, as shown in Figure 11.2.
- 7 The check-in detector should be located at the position where vehicles would normally stop at the stop line. A long detector (up to 4 m or longer) would be required to cover a wide range of stopping positions. This check-in detector is used to actuate the green light signal when an approaching vehicle is detected AND a minimum red period has expired.
- 8 The check-out detector actuates the red light signal as soon as a vehicle is detected, subject to the provision of a minimum green period (from the time the vehicle is first detected and NOT when it departs from the loop). The detector must be located beyond the last traffic signal at a point where the red light signal will not be visible to the departing vehicle (about 1 to 2 m beyond the signal).
- 9 Detector loops can also be installed on the freeway itself to ensure that an adequate gap exists before the next green light signal is provided on the ramp.
- 10 A queue detector can also be used to identify the backup of traffic onto the local street system. When a queue of that length is detected, a higher metering rate may be temporarily allowed to reduce the queue to an acceptable length. This must, however, be applied with circumspection because a consistently high rate of arrivals may result in the application of a metering rate that is so high that it approaches the condition of there being, in fact, no metering at all - thus defeating the objective of installing metering in the first instance.
- 11 The highest rate of metering that can be handled by metering of a single stream of vehicles is approximately 900 vehicles per hour. This rate can be attained with about two seconds of green followed by two seconds of red. A red period shorter than 2 seconds should not be used.
- 12 A lower rate of metering can be achieved by increasing the minimum red interval. A metering rate in the order of 600 vehicles per hour will, for instance, be achieved by providing two seconds of green followed by four seconds of red.

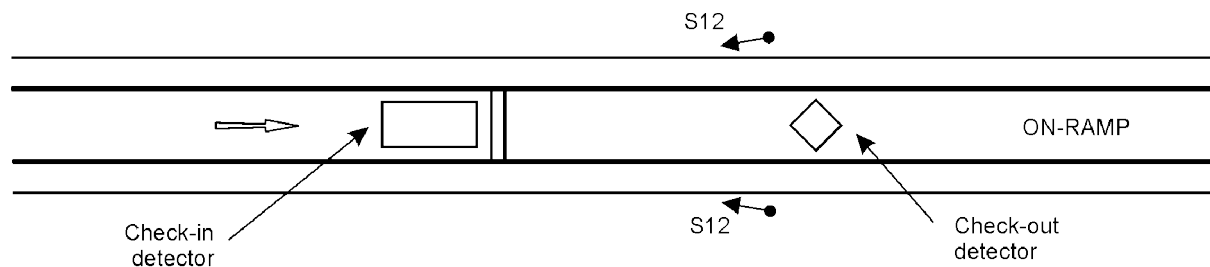


Figure 11.2: Ramp metering application of the S12 signal face